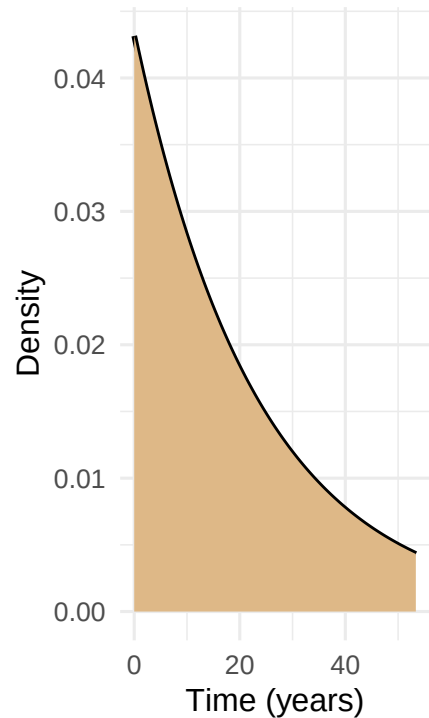
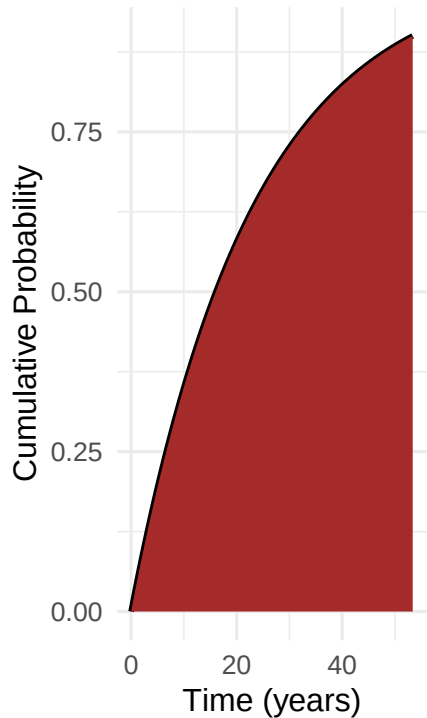


Solar Panel Reliability Report

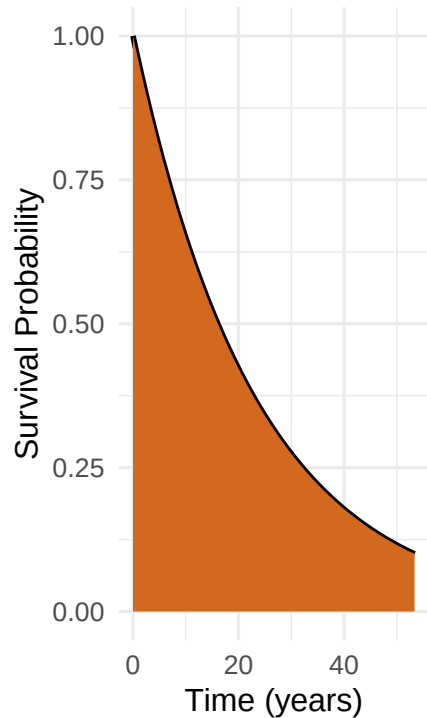
PDF



CDF



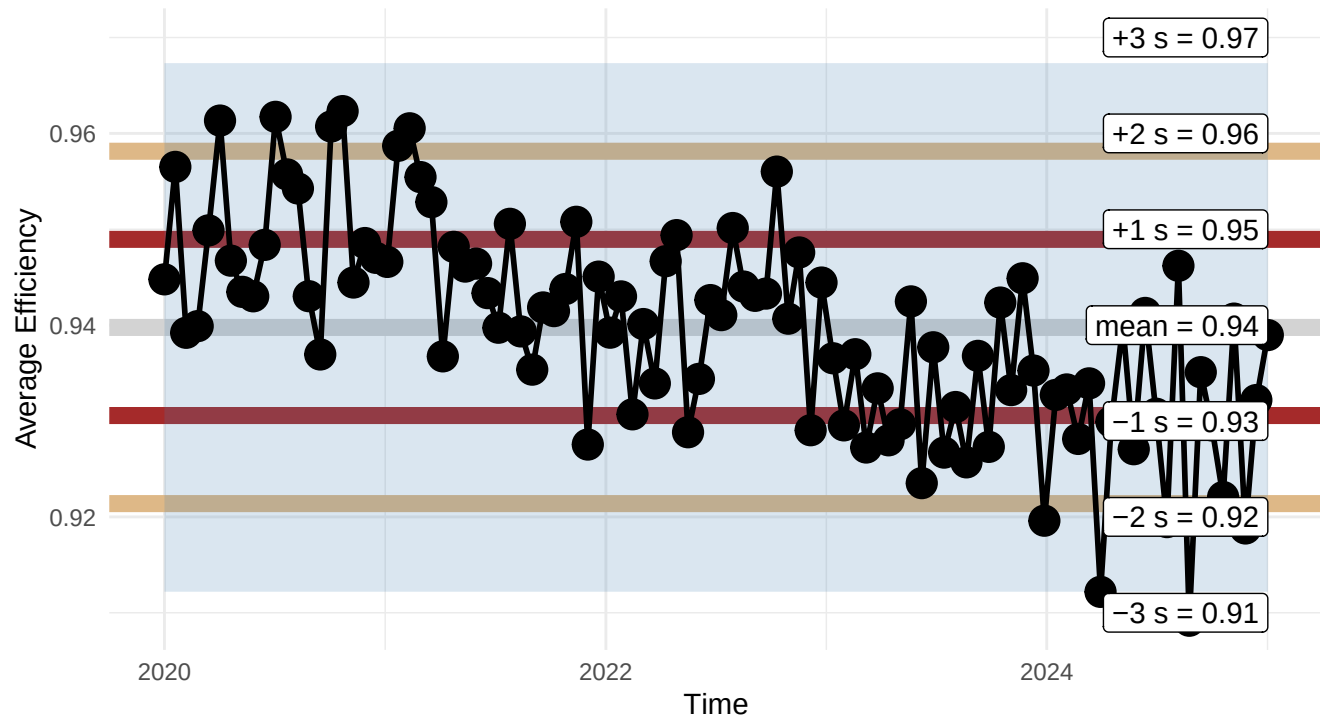
Survival Function



The solar panel farm's reliability was modeled using an exponential lifetime distribution, where the estimated mean time to failure was 23.2 years based on 100 panels and 21 observed failures. The model predicts that the expected survival rate is 65.0% after 10 years and 34.0% after 25 years. The control chart has a gradual decline or negative slope, suggesting decreasing average efficiency amongst the solar panels over time. Most of your failures are due to wiring issues. We recommend prioritizing electrical inspections and maintenance to prevent these failures.

Metric	Value
Total panels	100
Failed panels	21
MTTF (years)	23.2
Failure rate (/yr)	0.0432
Survival @10y	65.0%
Survival @20y	42.2%
Survival @25y	34.0%

Control Chart



Number of Solar Panels Failed by Failure Type

